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Abstract. abbbaaaaaaaa

1. Introduction

2. Model

3. Introduction and model formulation

Understanding the interaction between electrical excitation and recovery is a central problem in mathematical cardiac electrophysiology. Reduced models describe this interaction using a small number of variables, making it possible to investigate the mechanisms governing resting states, excitation thresholds, repolarization, and sustained oscillations. Their relatively simple structure also provides a setting in which qualitative properties can be established analytically.

Rogers and McCulloch [1] introduced a modification of FitzHugh–Nagumo-type kinetics in a computational study of cardiac action-potential propagation. Their formulation couples a voltage-dependent cubic excitation term to a recovery variable through a bilinear interaction. Although originally studied in a spatial propagation framework, its reaction terms also define a two-dimensional ordinary differential equation describing spatially homogeneous kinetics.

The present work concerns this autonomous planar system. The purpose is to investigate its qualitative dynamics as an established cardiac excitation model, rather than to introduce additional phenomenological interactions.

3.1. **The autonomous Rogers–McCulloch system.** Let V_m denote the transmembrane potential, and introduce the normalized voltage

$$u = \frac{V_m - V_r}{V_p - V_r}, \tag{1}$$

where V_r is a reference resting potential and $V_p > V_r$ is a reference excited potential. Let v be a dimensionless recovery variable representing the aggregate processes responsible for repolarization and recovery of excitability.

After absorbing the constant voltage and capacitance factors into the coefficients of the voltage equation, the spatially homogeneous Rogers–McCulloch kinetics can be written as

$$\begin{aligned} \frac{du}{dt} &= c_1 u(u-a)(1-u) - c_2 uv + I(t), \\ \frac{dv}{dt} &= b(u - dv). \end{aligned} \tag{2}$$

The parameter assumptions are

$$0 < a < 1, \qquad c_1, c_2, b, d > 0. \tag{3}$$

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Here t denotes time in the chosen physical unit. No state-dependent change of time is required to obtain the polynomial vector field.

To study autonomous phase portraits and limit cycles, we take a constant applied input,

$$I(t) \equiv I_0, \qquad I_0 \geq 0.$$

Thus, the system under consideration is

$$\boxed{\begin{aligned} \dot{u} &= c_1 u(u-a)(1-u) - c_2 uv + I_0, \\ \dot{v} &= b(u - dv). \end{aligned}} \tag{4}$$

A dot denotes differentiation with respect to t. System (4) is an autonomous planar polynomial system of degree three. The case $I_0 = 0$ describes unforced kinetics, whereas $I_0 > 0$ represents a sustained depolarizing input.

The coefficients in (4) refer to this normalized formulation. Numerical parameter values taken from a dimensional formulation must therefore be converted using the same voltage, current, capacitance, and time conventions.

3.2. **Interpretation of the variables and parameters.** The parameter a specifies the intermediate zero of the cubic excitation term. In voltage coordinates it corresponds to the reference level

$$V_{\mathrm{th}} = V_r + a(V_p - V_r).$$

It is a parameter controlling excitability, rather than a universal firing threshold independent of the recovery state and applied input.

The coefficient c_1 determines the strength of the intrinsic voltage kinetics. In the absence of recovery feedback and applied input, the cubic term is negative for $0 < u < a$, positive for $a < u < 1$, and negative for $u > 1$. This sign structure describes a competition between relaxation toward rest and regenerative excitation.

The coefficient c_2 determines the strength of the voltage–recovery interaction. For $u, v > 0$, the term $-c_2 uv$ opposes depolarization. Its dependence on both variables distinguishes this formulation from the usual additive recovery term in the standard FitzHugh–Nagumo equations.

The recovery equation can be expressed as

$$\dot{v} = bd\left(\frac{u}{d} - v\right). \tag{5}$$

Consequently, for a fixed voltage, the recovery variable relaxes toward u/d with relaxation time $(bd)^{-1}$. The parameter d determines the recovery nullcline, whereas b controls the recovery rate for a fixed value of d. A slow–fast interpretation requires an explicit comparison of this rate with the voltage kinetics; it does not follow from positivity of the parameters alone.

Finally, I_0 is the applied current expressed in normalized voltage-rate units. With dimensionless u and v, the parameters c_1, c_2, and b, together with I_0, have units of inverse time, while a and d are dimensionless.

The variable v is a phenomenological recovery variable. It should not be identified with a specific ionic concentration or individual channel-gating probability without additional justification. Similarly, the model describes electrical excitation and recovery, rather than the progression of a particular cardiac disease.

3.3. **Phase space and elementary geometric properties.** For nonnegative initial conditions, a natural domain is

$$\mathcal{Q} = \{(u, v) \in \mathbb{R}^2 : u \geq 0,\ v \geq 0\}.$$

Since

$$\dot{u}|_{u=0} = I_0 \geq 0, \qquad \dot{v}|_{v=0} = bu \geq 0,$$

the quadrant $\mathcal{Q}$ is positively invariant. When $I_0 = 0$, the boundary $u = 0$ is invariant; when $I_0 > 0$, the vector field points into $\mathcal{Q}$ along that boundary.

The normalization of voltage does not imply that $0 \le u \le 1$ is invariant under every applied input. In particular, at $u = 1$,

$$\dot{u} = I_0 - c_2 v,$$

which can be positive. Physiological interpretation must therefore account for the voltage range associated with a chosen parameter set and stimulation protocol.

Forward trajectories in $\mathcal{Q}$ are bounded. Indeed, because

$$c_1 u(u-a)(1-u) + I_0 \longrightarrow -\infty \quad \text{as } u \longrightarrow +\infty,$$

one can choose $U > 1$ such that this expression is negative for every $u \ge U$. Since $-c_2 uv \le 0$ in $\mathcal{Q}$,

$$u(t) \le M := \max\{u(0), U\}.$$

The recovery equation then gives

$$v(t) \le \max\left\{v(0), \frac{M}{d}\right\}.$$

These bounds also ensure forward completeness for nonnegative initial conditions.

The equilibria satisfy

$$v_* = \frac{u_*}{d}, \qquad c_1 u_*(u_* - a)(1 - u_*) - \frac{c_2}{d} u_*^2 + I_0 = 0. \tag{6}$$

Thus, there are at most three distinct finite equilibria. Their number, admissibility, and stability depend on the parameters.

A global analysis on $\mathcal{Q}$ concerns the dynamics of nonnegative states. A classification on the entire Poincare compactification is a broader mathematical problem and also includes regions without an immediate physiological interpretation. These two meanings of "global phase portrait" should be distinguished.

3.4. **Related work and research questions.** The original study of Rogers and McCulloch [1] emphasized numerical cardiac impulse propagation and the influence of tissue geometry and structure. More recently, López-Zazueta, Soto-Bajo, and Fraguela Collar [2] investigated numerically the response of an isolated cell under periodic pulse stimulation using the Rogers–McCulloch model.

At the analytical level, Hieber et al. [3] established the existence and uniqueness of strong time-periodic solutions near stable equilibria for periodically forced bidomain equations, including Rogers–McCulloch ionic transport.

These studies motivate, but do not by themselves resolve, the global classification problem for (4). A periodic response to a time-dependent stimulus is a different mathematical object from an isolated periodic orbit of an autonomous planar vector field. Likewise, a local existence theorem near a stable equilibrium does not determine the total number of autonomous limit cycles or the global configuration of separatrices.

The main questions for the autonomous system are therefore the following:

(1) How does parameter space divide according to the number and stability of equilibria, including nonhyperbolic configurations?
(2) For which parameter regimes do limit cycles exist or fail to exist? When they exist, what can be proved about their number, multiplicity, and stability?
(3) Which local and global bifurcations organize transitions between the admissible phase portraits, and which candidate bifurcations are excluded by the parameter restrictions?
(4) How do separatrices and basin boundaries organize resting, excited, and oscillatory behavior in $\mathcal{Q}$?

(5) If the analysis is extended to the whole plane, which singularities at infinity occur, and how do their separatrices connect with the finite dynamics?

These questions specify a research programme rather than a claim that each problem is absent from the existing literature. Establishing novelty requires comparison with previous analyses of the Rogers–McCulloch equations and with results for broader polynomial or generalized Liénard families that may contain equivalent systems.

In particular, the bilinear recovery coupling prevents a direct identification with the usual additive-recovery FitzHugh–Nagumo formulation. However, a different appearance of the equations is not sufficient evidence of mathematical novelty: coordinate transformations, time rescalings, and existing general theorems must also be examined.

A substantial contribution would consist of rigorous results beyond numerical examples or a routine local stability calculation, such as a sharp limit-cycle bound, a complete bifurcation classification on a specified parameter domain, or a global description of separatrix configurations and attraction basins. Any claim of completeness must specify the phase space, parameter domain, and treatment of degenerate cases.

Finally, autonomous oscillations in this reduced model describe repetitive electrical activity at the level of homogeneous kinetics. Their existence alone does not establish a mechanism for tissue-level reentry or a diagnosis of cardiac arrhythmia, which require additional spatial and physiological information.

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